

From Listeners to Creators: Bridging Musical Inequality through Generative AI Music Tools—A Practical Study Based on the ADDIE Model in Junior High School Education

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Background

Music is universally celebrated, yet composition has long remained accessible primarily to those with formal music theory training. Generative AI platforms such as SUNO AI have significantly lowered the technical threshold, shifting the creative barrier from technical mastery to conceptual expression. However, in junior high school settings, students with typing difficulties or limited language proficiency frequently struggle to translate internal emotions into precise AI prompts—a challenge this study terms the “Conceptual Materialization Dilemma.” Without deliberate pedagogical scaffolding, GenAI tools risk reproducing rather than dismantling existing inequalities in aesthetic education.

Objectives

This study aimed to design, implement, and evaluate a three-tiered cognitive scaffolding system enabling students with no formal music training to compose original multilingual songs using SUNO AI. Grounded in the ADDIE model and guided by Scaffolding Theory, Flow Theory, and Self-Determination Theory, the study sought to verify whether Human-AI collaboration can empower diverse learners to transcend linguistic barriers, regain creative agency, and actualize technological equity.

Methods / Implementation

A practitioner-research design was adopted across 18 heterogeneous classes (N = 496, Grades 7–8) at a public junior high school in Tainan, Taiwan, during May–June 2025. The six-week cross-disciplinary module integrated Taiwan’s Technology and Arts curriculum domains. The same cohort had prior AI scaffolding experience from the preceding semester. The researcher developed a three-tiered scaffolding system: (1) a Thematic Framework of 9 major themes and 180 sub-topics calibrated to adolescent emotional life; (2) a Prompt Scaffolding Website with visual drop-down menus enabling prompt generation without English typing; and (3) a Custom GPT model accepting Mandarin input and outputting professionally structured lyrics—with

automatic Mandarin translation and romanized phonetics—in any target language. Student outputs were archived in 18 class SUNO playlists and a Padlet digital gallery.

Results / Impact

The 496 student creators collectively produced 505 original compositions spanning Mandarin Chinese, English, Japanese, Korean, Vietnamese, Arabic, and additional languages, demonstrating that the scaffolding system effectively bridges virtually any linguistic divide. Qualitative observation across all 18 classes revealed sustained collective Flow states during Weeks 3–6, with behavioral disruptions virtually absent. Three students composed entirely self-authored lyrics, using the Custom GPT solely for structural formatting—evidence of genuine creative ownership cultivated by the scaffolding. Most compellingly, numerous students continued creating autonomously after the formal curriculum concluded, with several classes independently initiating graduation song projects recognized by faculty as artistically meritorious.

Discussion & Conclusion

This study demonstrates that GenAI tools, paired with intentionally designed cognitive scaffolding, can democratize aesthetic creation in junior high contexts. The educational divide in GenAI music creation resides not in hardware access but in cognitive and linguistic resources—addressable through Universal Design for Learning (UDL) principles aligned with SDG 4 and SDG 10. The ADDIE-grounded three-tiered scaffolding framework offers educators a replicable, tool-agnostic blueprint for large-scale AI integration. Building an intuitive, mobile-accessible interface requires design investment, and prior student AI literacy meaningfully accelerates adoption. Future research should incorporate validated instruments measuring flow, creative self-efficacy, and longitudinal autonomous learning behaviors.

Keywords

Generative AI, Music Education, Cognitive Scaffolding, ADDIE Model, Educational Equity